

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 20%

QUESTIONS: 05

Total Marks:50

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items (2)	Comparison Depth (2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)
Weightage	20%	25%	20%	20%	10%

Code of Conduct:

I KONGARA JOGESH(192525035) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:K.jogesh

RUBRICS FOR CALCULATION:

Comparative analysis

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

CSA16 – DATA WAREHOUSING AND DATA MINING

1. Introduction

Clustering is an unsupervised learning technique used to group observations that are similar according to selected features. In data warehousing and data mining, clustering can support customer segmentation, document organization, geographic grouping, product analysis and mixed-attribute discovery. The suitability of a clustering method depends on the structure of the data, the distance or similarity measure, the expected shape of groups, scalability requirements and the type of attributes available.

This assessment compares clustering methods using the five datasets specified in the question paper. Each answer contains: (1) the scope of the comparison, (2) a detailed comparison across multiple aspects, (3) evidence from the supplied values, (4) critical interpretation and (5) a concise recommendation. This structure directly addresses the Level 4 rubric requirements.

2. Q1 – Customer Data: K-means vs Hierarchical Clustering

Question: Compare K-means and hierarchical clustering using the customer dataset and explain differences in performance and scalability.

Customer	Income (₹)	Spending Score
C1	30,000	60
C2	60,000	40
C3	50,000	70
C4	28,000	65
C5	65,000	35

Customer Income vs Spending Score

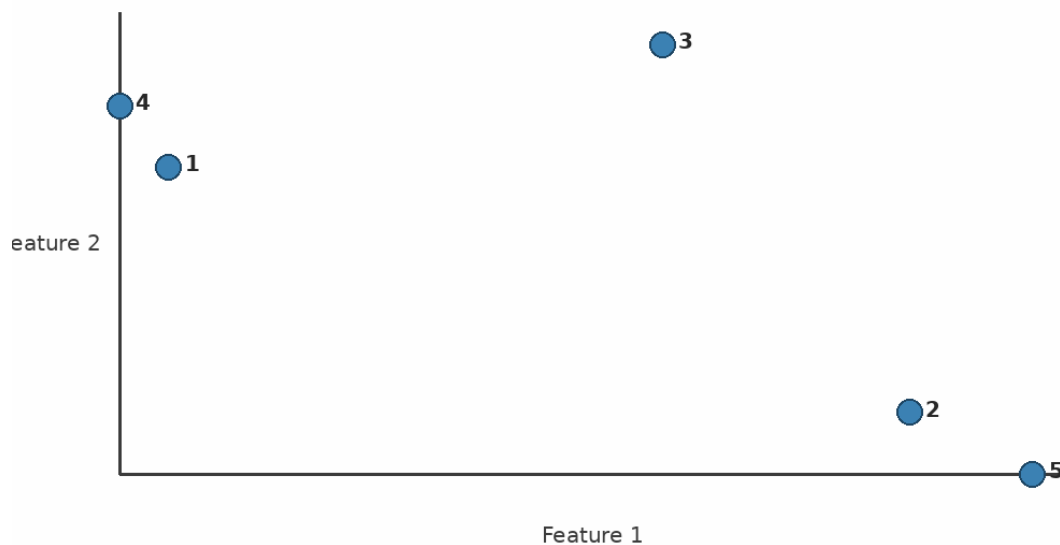


Figure 1. Customer income and spending-score distribution.

2.1 Scope of comparison

The comparison focuses on algorithmic mechanism, computational behaviour, scalability, interpretability and suitability for customer segmentation. These criteria are relevant because the question explicitly asks about performance and scalability.

2.2 K-means

K-means represents each cluster by a centroid. The algorithm alternates between assigning observations to the nearest centroid and recalculating the centroids until the assignments stabilize or a stopping condition is reached. It is particularly suitable for numerical features when clusters are reasonably compact and a target number of clusters can be specified.

2.3 Hierarchical clustering

Hierarchical clustering builds a hierarchy of groups. In the agglomerative form, each observation begins as its own cluster and similar clusters are progressively merged. The result can be represented using a dendrogram, allowing the analyst to examine cluster structure at different levels.

2.4 Evidence from the dataset

C2 and C5 have high income values of ₹60,000 and ₹65,000 with spending scores of 40 and 35. C1, C3 and C4 have lower incomes than C2/C5 and spending scores of 60, 70 and 65. This suggests a meaningful segmentation into a high-income/lower-spending group and a lower-to-moderate-income/higher-spending group.

2.5 Performance and scalability

- K-means is generally computationally efficient for large numerical datasets and can be rerun quickly when customer records increase.
- Hierarchical clustering is useful for exploratory analysis but becomes less attractive for very large datasets because maintaining and processing relationships between many observations is costly.
- K-means requires a choice of k , while hierarchical clustering can provide a complete hierarchy before a final cut is selected.
- For a small dataset such as the five records provided, both methods are practical; the scalability difference becomes more important as the customer base grows.

Critical interpretation: If the business needs repeated customer segmentation for a large database, K-means is the more scalable choice. If analysts need to explore nested relationships and do not want to commit immediately to one final number of clusters, hierarchical clustering provides greater exploratory flexibility.

Conclusion: K-means is preferable for scalable, repeated customer segmentation, while hierarchical clustering is valuable for exploratory analysis and visualizing relationships.

3. Q2 – Document Data: K-means vs EM for Overlapping Clusters

Question: Compare K-means and EM algorithms in handling overlapping clusters for the given document dataset.

Document	Word1	Word2	Word3
D1	.20	.50	.30
D2	.10	.60	.30

D3	.80	.10	.10
D4	.75	.15	.10
D5	.20	.40	.40

Document Feature Overlap

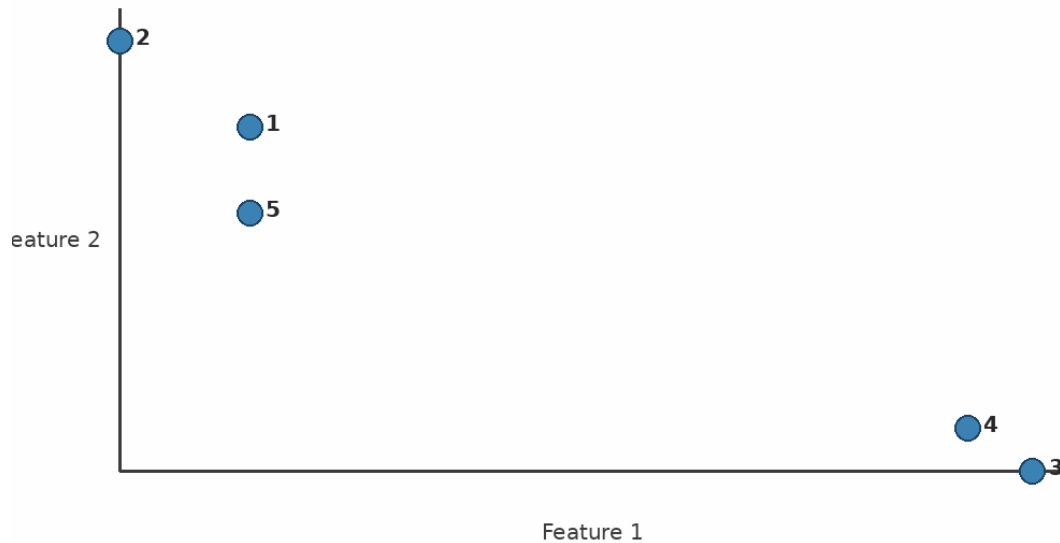


Figure 2. Document feature-space visualization.

3.1 Comparison

K-means produces hard assignments: each document is assigned to one cluster based on its distance from cluster centroids. This is simple and computationally efficient, but it can be restrictive when a document contains features associated with more than one topic.

Expectation-Maximization (EM), when applied with a probabilistic mixture model, estimates parameters and provides probabilistic membership. This makes EM useful when clusters overlap and an observation may reasonably belong to multiple groups with different probabilities.

3.2 Evidence and interpretation

D1 and D5 have similar Word1 values of 0.20 and Word2 values of 0.50 and 0.40. D3 and D4 are also close, with Word1 values of 0.80 and 0.75 and low Word2 values. D2 lies near the first group. This arrangement demonstrates why a hard boundary can hide uncertainty for observations near a cluster boundary.

3.3 Practical comparison

- K-means is easier to implement and explain when every document must receive one final label.
- EM is more expressive when documents can contain overlapping topics and the analyst wants membership probabilities.
- K-means usually requires the number of clusters to be specified; EM also requires model assumptions such as the form of the probability distribution.

- EM may require more computation and model-estimation effort, but the additional information can justify the cost when overlap is important.

Conclusion: EM is the stronger method for overlapping document clusters when soft membership is required; K-means is suitable when a simple hard partition is sufficient.

4. Q3 – Geographic Data: Euclidean vs Manhattan Distance

Question: Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results.

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

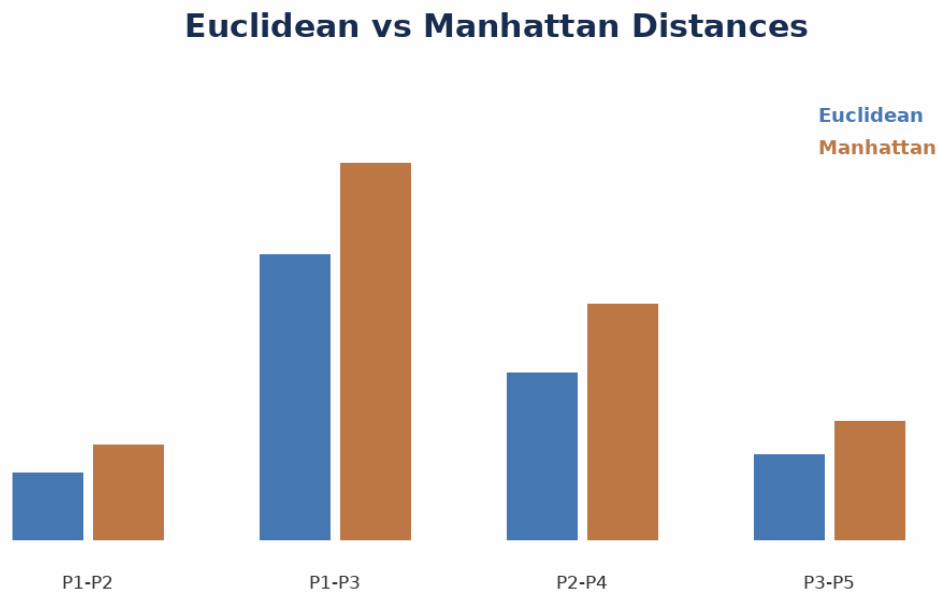


Figure 3. Selected Euclidean and Manhattan distance comparisons.

4.1 Definitions

Euclidean distance measures straight-line separation in coordinate space: $d = \sqrt{\sum (x_i - y_i)^2}$. Manhattan distance measures the sum of absolute coordinate differences: $d = \sum |x_i - y_i|$. The two metrics therefore impose different geometries on the same observations.

4.2 Evidence from the supplied points

P1–P2: Euclidean ≈ 0.0283 ; Manhattan = 0.0400.

P1–P3: Euclidean ≈ 0.1208 ; Manhattan = 0.1600.

P2–P4: Euclidean ≈ 0.0707 ; Manhattan = 0.1000.

P3–P5: Euclidean ≈ 0.0361 ; Manhattan = 0.0500.

4.3 Impact on clustering

- Changing the metric changes pairwise distances and therefore can change which points are considered nearest.
- Points near a cluster boundary can switch clusters when their relative distances change.
- Euclidean distance is natural for straight-line coordinate-space similarity, while Manhattan distance is useful when differences along separate dimensions are intended to accumulate independently.
- For real geographic systems, latitude/longitude usually requires a suitable geographic distance treatment; the present comparison follows the two metrics explicitly required by the assessment.

Conclusion: The distance metric is not a cosmetic choice; it influences similarity, cluster boundaries and final assignments. The metric should therefore match the interpretation of distance in the application.

5. Q4 – Retail Products: Agglomerative vs Divisive Clustering

Question: Compare agglomerative and divisive clustering approaches and explain their suitability for the retail dataset.

Product	Sales (₹)	Frequency
P1	50,000	20
P2	30,000	10
P3	52,000	22
P4	25,000	8
P5	32,000	12

Retail Products - Agglomerative Hierarchy

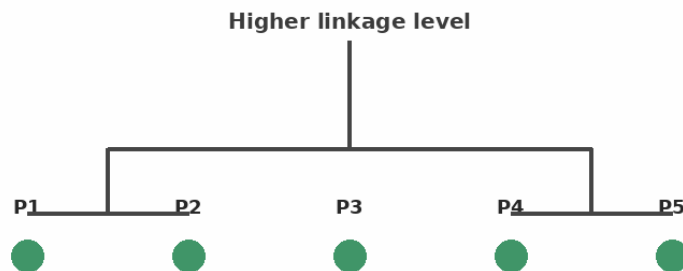


Figure 4. Illustrative hierarchy for the retail products.

5.1 Agglomerative approach

Agglomerative clustering starts with each product as an individual cluster and repeatedly merges the most similar clusters. The process creates a hierarchy that can be inspected at different levels, making it useful for exploratory retail analysis.

5.2 Divisive approach

Divisive clustering starts with all products in a single cluster and recursively divides the cluster into smaller groups. This top-down structure can be attractive when a business wants to begin with the complete product population and progressively create more specific segments.

5.3 Evidence from the data

P3 has the highest sales (₹52,000) and frequency (22), followed by P1 with ₹50,000 and frequency 20. P4 is lowest on both measures at ₹25,000 and frequency 8. P2 and P5 occupy an intermediate/lower region. These values provide a clear basis for grouping products by sales activity.

5.4 Suitability

- Agglomerative clustering is convenient for a small dataset because the analyst can examine the complete hierarchy and select a useful level of segmentation.
- Divisive clustering is useful when the business question naturally starts with one market and asks how it can be split into meaningful subgroups.
- Both methods can be affected by the selected distance measure and linkage/splitting strategy.
- For the five-product dataset, the transparency of an agglomerative hierarchy makes it a strong practical choice for exploratory analysis.

Conclusion: Agglomerative clustering is well suited to this small retail dataset because it provides an interpretable hierarchy of product similarity; divisive clustering remains useful for top-down segmentation.

6. Q5 – Mixed Attributes: Centroid-Based vs Density-Based Clustering

Question: Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types.

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Mixed Attributes - Price and Rating

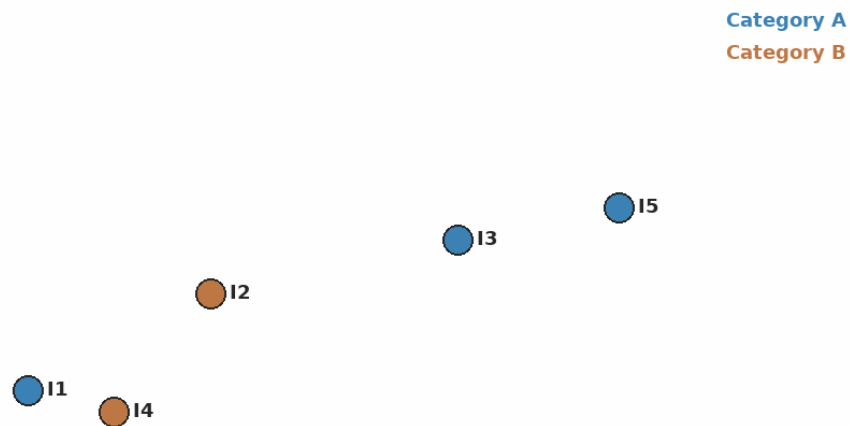


Figure 5. Mixed-attribute visualization using price, rating and category.

6.1 Centroid-based techniques

Centroid-based clustering, represented by K-means, summarizes each cluster using a centroid and assigns observations according to distance. It works naturally with numerical attributes after suitable scaling. Categorical attributes such as Category A/B cannot simply be treated as ordinary continuous numbers because doing so can create an artificial numeric relationship.

6.2 Density-based techniques

Density-based clustering, such as DBSCAN, identifies dense regions separated by areas of lower density and can label some observations as noise. This can be valuable when clusters are irregularly shaped or when outlier detection is important.

6.3 Evidence from the dataset

Category A items I1, I3 and I5 have prices ₹500, ₹550 and ₹520 with ratings 4.5, 4.7 and 4.6. Category B items I2 and I4 have lower prices of ₹300 and ₹250 and ratings 3.8 and 3.5. The visual separation provides evidence that category is associated with the numeric attributes in this small dataset.

6.4 Mixed-data challenge and critical thinking

- Direct K-means on raw mixed data is inappropriate because the categorical variable does not have a meaningful continuous scale.
- One option is suitable categorical encoding combined with scaling, but the encoding must not introduce misleading distance relationships.
- Another option is a clustering approach and distance measure specifically designed for mixed numerical and categorical attributes.

- Density-based methods can detect dense groups and noise, but their effectiveness depends on parameter choices and a meaningful distance definition.
- Because the dataset is very small, the main challenge is correct representation of the mixed attributes rather than computational scalability.

Conclusion: For mixed data, algorithm selection must follow correct attribute representation. A centroid-based approach can be effective after appropriate preprocessing, while density-based clustering is useful when discovering dense groups and possible outliers is the priority.

7. Cross-Question Comparative Discussion

The five questions illustrate different dimensions of clustering applicability. The customer problem emphasizes scalability; the document problem emphasizes overlapping membership; the geographic problem emphasizes distance geometry; the retail problem emphasizes hierarchical construction; and the mixed-attribute problem emphasizes data representation.

Scenario	Main Issue	Strong Candidate	Reason
Customer segmentation	Performance & scalability	K-means	Efficient numeric segmentation for growing datasets
Document clustering	Overlapping groups	EM	Probabilistic membership represents uncertainty
Geographic points	Distance definition	Depends on metric	Metric changes similarity and cluster boundaries
Retail products	Hierarchy	Agglomerative	Interpretable bottom-up hierarchy
Mixed attributes	Data type compatibility	Preprocessed centroid or suitable density method	Requires correct treatment of categorical + numeric data

8. Data Preprocessing and Validation Considerations

- Scale numeric variables when their ranges differ substantially so one feature does not dominate distance calculations.
- Check for missing values, duplicate records and outliers before clustering.
- Choose the distance/similarity measure according to the data type and application meaning.
- For K-means, evaluate a suitable value of k rather than selecting it arbitrarily.
- For hierarchical clustering, inspect the dendrogram and compare meaningful cut levels.
- For density-based clustering, select parameters carefully and validate whether identified noise points are genuine anomalies.
- Validate clusters using both quantitative measures and domain interpretation; a mathematically compact cluster is not automatically a meaningful business segment.

9. Rubric-Based Assessment Alignment

The source rubric assigns Scope of Items 20%, Comparison Depth 25%, Use of Evidence 20%, Critical Thinking 20% and Clarity of Presentation 15%. The report has been expanded specifically to meet the Level 4 descriptors.

Criterion	Weightage	How the assessment addresses Level 4
Scope of Items	20%	Each answer uses the exact dataset and

		comparison requested; the scope is stated and justified before analysis.
Comparison Depth	25%	Each question compares multiple aspects such as mechanism, scalability, assumptions, interpretation, strengths, limitations and practical suitability.
Use of Evidence	20%	The supplied values are reproduced in tables, pairwise distance calculations are shown, and five figures visualize the datasets.
Critical Thinking	20%	The report interprets the evidence, discusses limitations and trade-offs, and gives a justified recommendation for each scenario.
Clarity of Presentation	15%	The report is logically structured with editable front pages, rubric, numbered sections, tables, figures, evidence and conclusions.

10. Overall Conclusion

The analysis demonstrates that no single clustering method is universally superior. K-means is attractive for scalable numerical segmentation; hierarchical clustering is valuable for exploratory structure; EM is useful when membership can overlap; distance metrics directly affect geographic clustering; and mixed-attribute clustering requires careful preprocessing or a method that correctly represents categorical information. Therefore, the most appropriate algorithm should be selected according to the structure of the dataset, the intended interpretation of similarity, computational requirements and the practical objective of the analysis.